

FIT2179 Data Visualisation 2

Australia's Olympic & Paralympic Legacy

Introduction

Australia has long punched well above its weight on the Olympic stage, a remarkable achievement for a country of just 26 million people. Yet behind the medals sits a rich, data-driven story that most Australians have never seen told in a visual, accessible way.

This visualisation explores Australia's 130-year Olympic and Paralympic journey: the nations we outperform, the sports that define us, the home-games effect, the rise of our Paralympic athletes, and where we stand today after the Paris 2024 Games. Every chart is grounded in real, publicly available data and designed to be understood by any Australian reader.

The story is structured across five sections, moving from the big global picture down to individual sports, then outward again to the future of the Brisbane 2032 Games. Scroll through, explore, and discover just how extraordinary Australia's Olympic legacy truly is.

Section 1: Australia on the World Stage

Australia has won more than 500 Summer Olympic medals since its first Games in 1896. Measured per million people, it outperforms the United States, China, and most of Europe.

Chart A — “Small country, big medals”: World Medal Map

Field	Detail
Idiom	Choropleth Map (REQUIRED MAP)
What it shows	A world map shaded by total Olympic medals won per million people (per capita). Australia lights up dramatically compared to similarly sized nations, immediately showing the viewer that Australia punches well above its weight on the global stage.
Data source 1	Kaggle — 120 Years of Olympic History (Athletes and Results): https://www.kaggle.com/datasets/heesoo37/120-years-of-olympic-history-athletes-and-results
Data source 2	World Bank Population Data: https://data.worldbank.org/indicator/SP.POP.TOTL
How to combine	Join Olympic medals dataset with World Bank population figures to derive ‘medals per million’ — this satisfies the assignment’s two-source combination requirement.

Chart B — Top 15 Nations: Medals Per Million People

Field	Detail
Idiom	Horizontal Bar Chart (sorted, with Australia highlighted in gold)
What it shows	A sorted horizontal bar chart of the top 15 nations ranked by medals per million population. This directly complements the map by giving exact numbers and a ranking, making the comparison concrete and easy to read.
Data source	Kaggle — 120 Years of Olympic History. Aggregate total medals per country, then divide by World Bank population figures to compute per-capita rank.

Chart C — Australia’s Medal Trajectory, 1896–2024

Field	Detail
Idiom	Multi-series Line Chart with annotations
What it shows	Three lines (Gold, Silver, Bronze) showing Australia’s medal count across every Summer Games from 1896 to Paris 2024. Key moments annotated directly on the chart: the 1956 Melbourne Games, the 2000 Sydney home Games spike, and the 2021 Tokyo Games. Interactive tooltip reveals the host city and exact medal count on hover.
Data source	Kaggle — 120 Years of Olympic History. Filter rows to Australia (NOC = ‘AUS’), group by Year and Medal type, count rows per group.

Section 2: Where Australia Wins — and Where It Doesn’t

Not all sports are created equal in Australia. Aquatics alone accounts for nearly a third of all Australian Olympic gold medals. Meanwhile, Australia has won almost nothing in the Winter Games, reflecting both geography and culture.

Chart A — “Sports That Define Us”: Gold Medal Unit Chart

Field	Detail
Idiom	Waffle / Unit Chart (HD custom idiom)
What it shows	Every Australian gold medal is represented as one small square, grouped by sport category (Aquatics, Athletics, Cycling, Rowing, etc.). The viewer instantly sees that Aquatics dominates. Far more engaging and memorable than a pie chart.
Data source	Kaggle — 120 Years of Olympic History. Filter to AUS + Gold medals, group by Sport.
How to build	In Vega-Lite, use a point/square mark with calculated x/y positions derived from a running count. This is the ‘derived data’ the HD rubric specifically rewards.

Chart B — Summer vs Winter: The Gap

Field	Detail
Idiom	Stacked Bar Chart
What it shows	Side-by-side bars for Summer and Winter Games across decades, showing total medals in each. Australia's near-zero Winter Games presence compared to Summer dominance tells an interesting cultural story about geography and tradition.
Data source	Kaggle — 120 Years of Olympic History. Group by Season (Summer/Winter) and decade, count total medals. The dataset already includes a Season column.

Chart C — Australia's Sport Performance Heatmap by Decade

Field	Detail
Idiom	Heatmap Matrix (HD custom idiom)
What it shows	Rows are sports (top 15 by total medals), columns are Olympic decades (1900s–2020s). Cell colour intensity encodes medal count. Immediately shows swimming's long unbroken dominance, and newer strengths like cycling and BMX emerging after 2000.
Data source	Kaggle — 120 Years of Olympic History. Filter to AUS, group by Sport and decade ($\text{floor}(\text{Year}/10) * 10$), count medals. In Vega-Lite, use rect mark with colour encoding on count.

Section 3: The Home Games Effect — Sydney 2000

Sydney 2000 remains the high-water mark of Australian Olympic achievement. Australia finished 4th on the overall medal tally, extraordinary for a country of its size. Data can prove what memory already tells us: hosting the Games genuinely changes outcomes.

Chart A — Sydney 2000 vs Our Average: Sport-by-Sport Comparison

Field	Detail
Idiom	Lollipop / Dot Plot with deviation line
What it shows	Each sport on the y-axis. Two dots per sport: the Sydney 2000 medal count and the 5-Games average (1992–2012). A horizontal line connects them. Sports with large positive gaps (Swimming, Athletics, Rowing) make the home advantage effect visually obvious.
Data source	Kaggle — 120 Years of Olympic History. Filter to AUS. For the 5-Games average: filter to years 1992, 1996, 2000, 2004, 2008. For Sydney 2000 specifically: filter Year = 2000. Group both by Sport, count medals, then subtract to get deviation.

Chart B — Where Did Australia's Medals Come From? State Map

Field	Detail
Idiom	Australian State Choropleth Map
What it shows	A map of Australia's states and territories shaded by the number of total Olympic medals won by athletes from each state across all Games (based on state of the athletes' representing club/state). Reveals whether smaller states punch above their weight relative to population.
Data source 1	Kaggle — 120 Years of Olympic History (NOC = AUS, with team/state field where available).
Data source 2	ABS ASGS State Boundaries GeoJSON for the map layer: https://www.abs.gov.au/statistics/standards/australian-statistical-geography-standard-asgs-edition-3/jul2021-jun2026/access-and-downloads/digital-boundary-files
Note	If per-athlete state data is sparse in Kaggle, use AIS (Australian Institute of Sport) state centre data as a proxy — AIS publishes state-by-state athlete support numbers which is a reliable substitute.

Section 4: Paralympic Pride — The Overlooked Legacy

Australia has won over 1,000 Paralympic medals since the Games began. In Tokyo 2020, it finished 6th on the Paralympic medal tally. Yet this extraordinary record is rarely discussed. The data tells a story of resilience, investment, and national pride that deserves to sit alongside the Olympic story.

Chart A — Paralympic Medals: Australia vs the World

Field	Detail
Idiom	Grouped Bar Chart
What it shows	Australia alongside the top 5 Paralympic nations (USA, China, Great Britain, Germany, Canada) across Summer Paralympic Games since 1988. Demonstrates that Australia's Paralympic performance mirrors its Olympic strength — consistently elite despite its small population.
Data source	International Paralympic Committee (IPC) — Historical Results: https://www.paralympic.org/results/historical . Download medal tables by country and Games from the IPC website directly.

Chart B — Olympic vs Paralympic Rank: Do They Move Together?

Field	Detail
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Idiom	Connected Dot Plot / Slope Chart (HD custom idiom)
What it shows	For each Games year since 1988, two dots are plotted: Australia’s final-medal-tally rank in the Olympics and its rank in the Paralympics. The dots are connected by a line. If Australia performs relatively better in one than the other, the line slopes — making a pattern immediately visible.
Data source 1	Kaggle — 120 Years of Olympic History (aggregate all nations’ totals per Games, rank Australia).
Data source 2	IPC Historical Data — derive Australia’s Paralympic rank per Games. Join both by year.

Section 5: Australia’s Future — Paris 2024 & Beyond

Paris 2024 saw Australia win 53 medals, its best overseas performance ever. With Brisbane 2032 confirmed as the next home Games, history suggests Australia is positioned for another Sydney 2000-style surge. The data gives us reason to be genuinely excited about what comes next.

Chart A — Paris 2024: Australia’s Medal Breakdown by Sport

Field	Detail
Idiom	Annotated Stacked Bar Chart (Gold / Silver / Bronze by sport)
What it shows	Sport-by-sport medal breakdown for the Paris 2024 Games. Bars coloured by Gold/Silver/Bronze. Annotations call out standout performances (Swimming, BMX, Sailing). Grounds the narrative in the present before the forward-looking close.
Data source	Olympics.com — Paris 2024 Official Medal Data: https://olympics.com/en/olympic-games/paris-2024/medals/australia . Cross-check with the Australian Olympic Committee: https://olympics.com.au/games/paris-2024/

Chart B — Age of Australia’s Medal Winners Across the Decades

Field	Detail
Idiom	Box Plot (distribution per decade)
What it shows	One box plot per Olympic decade showing the age distribution of Australian medal-winning athletes. Reveals whether Australia’s elite athletes are getting older or younger over time — a subtle story about talent pipelines and sports science improvements.
Data source	Kaggle — 120 Years of Olympic History. Filter to AUS medal winners only. The dataset includes athlete Age at time of Games. Group by decade ($\text{floor}(\text{Year}/10) * 10$). In Vega-Lite, use the boxplot mark.
Note	The Kaggle dataset has some missing age values — filter these out and note the caveat in your annotation.

Chart C — Brisbane 2032 Projection: Will History Repeat?

Field	Detail
Idiom	Annotated Line Chart with projected range
What it shows	Australia's total medal count at each Games, with a shaded projection band for Brisbane 2032 based on the home-Games uplift seen at Melbourne 1956 and Sydney 2000. Ends the visualisation on a forward-looking, optimistic note grounded in historical evidence.
Data source	Kaggle — 120 Years of Olympic History (historical trend). The projection band is calculated as current trend +/- the average home-Games uplift across Melbourne 1956 and Sydney 2000 (derived data). Annotate Brisbane 2032 clearly as a projection, not a prediction.

Conclusion

Australia's Olympic and Paralympic story is, at its core, a story about identity. A young, sport-obsessed nation that has turned a passion for competition into one of the most impressive per-capita sporting records in history. From the swimmer who wins gold to the Paralympic athlete breaking world records, the data consistently tells the same story: Australia shows up, and Australia performs.

This visualisation has moved through five chapters of that story: global dominance, sporting strengths, the magic of home Games, Paralympic excellence, and a future that looks brighter than ever.

With Brisbane 2032 on the horizon, the next chapter of this story is still being written. The data gives us every reason to believe it will be the best one yet.

Data Sources Summary

All data sources are publicly accessible and credible:

- Kaggle — 120 Years of Olympic History (derived from official IOC/Olympic Games archives): <https://www.kaggle.com/datasets/heesoo37/120-years-of-olympic-history-athletes-and-results>
- World Bank — Population Data: <https://data.worldbank.org/indicator/SP.POP.TOTL>
- International Paralympic Committee (IPC) — Historical Results: <https://www.paralympic.org/results/historical>
- Olympics.com — Paris 2024 Medal Data: <https://olympics.com/en/olympic-games/paris-2024/medals/australia>
- ABS — ASGS State Boundaries GeoJSON: <https://www.abs.gov.au/statistics/standards/australian-statistical-geography-standard-asgs-edition-3/jul2021-jun2026/access-and-downloads/digital-boundary-files>